Yulong Shi

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Education

Northeastern University 09/2024 – 06/2027

M.S. in Engineering, Biomedical Engineering

Guilin University of Electronic Technology 09/2020 – 07/2024

B.S. in Engineering, Measurement and Control Technology and Instrument

Publications

Published

1. Tell2Adapt: A Unified Framework for Source Free Unsupervised Domain Adaptation via Vision Foundation Model
   IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR 2026) **CCF-A, CORE-A*** First author
2. Search2Adapt: Multi-Agent Collaborative Tree Search for Source-Free Unsupervised Domain Adaptation
   34th ACM International Conference on Multimedia (ACM MM 2026) **CCF-A, CORE-A*** First author
3. HEAL: Learning-Free Source Free Unsupervised Domain Adaptation for Cross-Modality Medical Image Segmentation
   The 36th British Machine Vision Conference (BMVC 2025) **CCF-C, CORE-A** First author
4. A^3-DualUD: Source-Free Unsupervised Domain Adaptation via Anatomical Anchor Alignment and Dual-path Uncertainty Denoising for Cross-modality Medical Image Segmentation
   Computer Methods and Programs in Biomedicine **IF 4.8, JCR Q1** First author

Under Review

1. PARDA: A Perceive-Act-Reflect-Distill Agent for Source-Free Unsupervised Domain Adaptation
   IEEE Transactions on Medical Imaging **IF 13.0, JCR Q1** First author
2. Source-free Unsupervised Domain Adaptation with Channel-Masked Feature Refinement and Memory Augmented Prototype Contrast for Cross-modality Medical Image Segmentation
   Biomedical Signal Processing and Control **IF 5.1, JCR Q1** First author
3. Source-Free Unsupervised Domain Adaptation for Medical Image Analysis: Progress, Challenges, and Future Prospects
   Artificial Intelligence In Medicine **IF 6.9, JCR Q1** First author

Internship Experience

Huawei Intelligent Automotive Solution BU Multimodal Foundation Model Engineer 06/2026 – 09/2026

Proposed RiskTrace-RL, a risk-aware RL post-training framework for autonomous-driving world-action models. Formulated the generative denoising trajectory as an MDP to address decision-information loss caused by prematurely passing noisy video representations to the action expert, and developed a task-specific reward model integrating temporal latents, handoff K/V features, action trajectories, and multidimensional safety signals. Applied safety-constrained, world-model-only policy optimization with cross-modal causal ablations to mitigate reward hacking, improving closed-loop NAVSIM PDMS to 91.8%.

Research Experience

Perceive-Act-Reflect-Distill Agent for Source-Free Unsupervised Domain Adaptation

Propose a novel framework that reformulates SFUDA as a closed-loop decision-making process. Developed a VLM-driven agent to analyze target-domain features and coordinate BiomedParse and SAM3 for high-quality pseudo-label generation. Introduced a Perceive-Act-Reflect-Distill paradigm with anatomical reliability assessment and iterative correction, effectively reducing error accumulation in conventional open-loop methods. This work is major revision at IEEE TMI.

Multi-Agent Collaborative Tree Search for Source-Free Unsupervised Domain Adaptation

Develope a collaborative multi-agent SFUDA framework consisting of Perceptor, Strategist, Executor, and Evaluator. Designed a collaborative tree-based search strategy to explore optimal domain adaptation plans within a complex operator space. Proposed a multimodal, anatomically-aware reward mechanism that integrates geometric textual descriptions with visual cues, enabling closed-loop calibration between Strategist and Evaluator. The method significantly improves search stability and pseudo-label quality for cross-modality segmentation. This work is under review at ACM MM 2026.

Scholarship and Awards

National Scholarship	10/2025
First Prize of 10th National Biomedical Engineering Innovation Design Competition for College Students	07/2025
Outstanding Graduates of Guangxi Autonomous Region	05/2024
National Encouragement Scholarship	11/2022